THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A Maya numeral has 5 in the 20s place and 7 in the units place. Its value is: 1

(a) 93 **(b)** 107 **(c)** 57 **(d)** 12

2. Which Roman symbols are never repeated? 1

(a) I, V and X **(b)** I, X and C **(c)** V, L and D **(d)** C, D and M

3. Which of the following is NOT a correct Roman numeral? 1

(a) IL **(b)** VI **(c)** XIX **(d)** XIV

4. The decimal value of 1201_3 is: 1

(a) 46 **(b)** 48 **(c)** 51 **(d)** 44

5. The number 50 written in base 3 is: 1

(a) 1212 **(b)** 2121 **(c)** 1211 **(d)** 1220

6. A Babylonian numeral has 1 group(s) of 60 and 25 units. Its value is: 1

(a) 26 **(b)** 85 **(c)** 31 **(d)** 35

7. A Maya numeral has 11 in the 20s place and 4 in the units place. Its value is: 1

(a) 216 **(b)** 15 **(c)** 224 **(d)** 114

8. The Roman symbol I can be repeated at most: 1

(a) 5 times **(b)** 3 times **(c)** 2 times **(d)** 4 times

9. Which of the following is NOT a correct Roman numeral? 1

(a) XM **(b)** XLIV **(c)** IX **(d)** IV

10. The decimal value of 3021_4 is: 1

(a) 201 **(b)** 199 **(c)** 206 **(d)** 203

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write 37 in the binary system. 2

12. A Babylonian numeral is made of 2 sixties and 48 units, and a Maya numeral has 6 twenties and 9 units. Find their values. 2

13. Convert 2102_3 to a decimal number. 2

14. Write 45 in the binary system. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Convert 89 to base 3 and check your answer by converting back to decimal. 3

16. Find CLVI + CCXLIX and write the answer in Roman numerals. 3

17. Write 21012_3 in expanded form and find its decimal value. 3

OR

Convert 175 to base 6 and check your answer by converting back to decimal. 3

18. Find CCLXXV + CCCXVIII and write the answer in Roman numerals. 3

OR

Write 5043_6 in expanded form and find its decimal value. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Add the binary numbers 10011_2 and 1011_2. Verify your answer by converting to decimal. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 27 kg using these weights, each at most once. Explain the idea using binary numbers. 5

20. Add the binary numbers 11110_2 and 10101_2. Verify your answer by converting to decimal. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 35 kg using these weights, each at most once. Explain the idea using binary numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

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